

Deviance, AIC, and model building

SBA data

Data on loan defaults for US Small Business Administration (SBA) loans:

- + Default: whether the business defaulted on the loan (1 = yes, 0 = no)
- + UrbanRural: 1 if business is in urban area, 2 if business is in rural area, 0 if unknown
- + NewExist: 1 if business already existed, 2 if business is new, 0 if unknown
- + Term: Length of the loan term (months)
- + DisbursementGross: The amount of money disbursed (loaned), in dollars
- + Many other variables...

Last time: Class activity

```
m1 <- glm(Default ~ log(DisbursementGross) + Term +  
          as.factor(UrbanRural),  
          family = binomial, data = sba)  
  
table(Prediction = m1$fitted.values > 0.5,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3989  734  
##      TRUE   100  168
```

- + Accuracy =
- + Sensitivity =
- + Specificity =
- + PPV =

Last time: Class activity

##		Truth	
##	Prediction	FALSE	TRUE
##	FALSE	3989	734
##	TRUE	100	168

Is an accuracy of around 80% good?

Last time: Class activity

Changing thresholds:

```
table(Prediction = m1$fitted.values > 0.3,  
      Truth = sba$Default)
```

```
##              Truth  
## Prediction FALSE TRUE  
##      FALSE  3524  351  
##      TRUE   565  551
```

```
table(Prediction = m1$fitted.values > 0.7,  
      Truth = sba$Default)
```

```
##              Truth  
## Prediction FALSE TRUE  
##      FALSE  4089  902
```

A new research question

- + Default: whether the business defaulted on the loan (1 = yes, 0 = no)
- + UrbanRural: 1 if business is in urban area, 2 if business is in rural area, 0 if unknown
- + NewExist: 1 if business already existed, 2 if business is new, 0 if unknown
- + Term: Length of the loan term (months)
- + DisbursementGross: The amount of money disbursed (loaned), in dollars
- + Many other variables...

Research question: Which combination of variables "best" models loan default?

A new research question

Research question: Which combination of variables "best" models loan default?

We need:

- + A metric to compare different models
- + A way to efficiently search through many different models

Logistic regression

Data: Grad application data

- + admit: accepted to grad school? (0 = no, 1 = yes)
- + gre: GRE score
- + gpa: undergrad GPA
- + rank: prestige of undergrad institution

$$Admit_i \sim Bernoulli(\pi_i) \quad \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Goal: Summarize how well our model fits the data

How do we summarize model fit in *linear* regression?

Summarizing linear regression model fit

Example model:

```
linear_mod <- lm(gre ~ gpa, data = grad_app)
summary(linear_mod)
```

```
...
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   192.30      47.92    4.013 7.15e-05 ***
## gpa           116.64      14.05    8.304 1.60e-15 ***
## Residual standard error: 106.8 on 398 degrees of freedom
## Multiple R-squared:  0.1477,    Adjusted R-squared:  0.1455
...
```

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## Multiple R-squared:  0.1477,    Adjusted R-squared:  0.1455
...
```

Is R^2_{adj} appropriate for logistic regression?

Summarizing logistic regression

```
logistic_mod <- glm(admit ~ gpa, data = grad_app,  
                    family = binomial)  
summary(logistic_mod)
```

```
...  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)  -4.3576      1.0353  -4.209 2.57e-05 ***  
## gpa          1.0511      0.2989   3.517 0.000437 ***  
## ---  
##      Null deviance: 499.98  on 399  degrees of freedom  
## Residual deviance: 486.97  on 398  degrees of freedom  
...
```

Deviance

```
logistic_mod <- glm(admit ~ gpa, data = grad_app,  
                    family = binomial)  
summary(logistic_mod)
```

```
...  
##              Estimate Std. Error z value Pr(>|z|)  
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## ---  
##      Null deviance: 499.98  on 399  degrees of freedom  
## Residual deviance: 486.97  on 398  degrees of freedom  
...
```

Deviance

Example:

```
glm(admit ~ gre, family = binomial, data = grad_app)
```

```
...  
## Degrees of Freedom: 399 Total (i.e. Null); 398 Residual  
## Null Deviance: 500  
## Residual Deviance: 486.1 AIC: 490.1  
...
```

```
glm(admit ~ gpa, family = binomial, data = grad_app)
```

```
...  
## Degrees of Freedom: 399 Total (i.e. Null); 398 Residual  
## Null Deviance: 500  
## Residual Deviance: 487 AIC: 491  
...
```

Which predictor (GRE or GPA) gives a model with a better fit?

Comparing models

```
glm(admit ~ gre, family = binomial, data = grad_app)
```

```
...
```

```
## Degrees of Freedom: 399 Total (i.e. Null); 398 Residual
```

```
## Null Deviance: 500
```

```
## Residual Deviance: 486.1 AIC: 490.1
```

```
...
```

```
glm(admit ~ gre + gpa, family = binomial, data = grad_app)
```

How will deviance change when I add another variable to the model?

Comparing models

```
glm(admit ~ gre, family = binomial, data = grad_app)
```

```
...
```

```
## Degrees of Freedom: 399 Total (i.e. Null); 398 Residual
```

```
## Null Deviance: 500
```

```
## Residual Deviance: 486.1 AIC: 490.1
```

```
...
```

```
glm(admit ~ gre + gpa, family = binomial, data = grad_app)
```

```
...
```

```
## Degrees of Freedom: 399 Total (i.e. Null); 397 Residual
```

```
## Null Deviance: 500
```

```
## Residual Deviance: 480.3 AIC: 486.3
```

```
...
```

Deviance will *never* increase when adding more explanatory variables to a model!

Comparing models

In linear regression, what quantity did we use to compare models with different numbers of parameters?

AIC

```
glm(admit ~ gre + gpa, family = binomial, data = grad_app)
```

```
...
```

```
## Degrees of Freedom: 399 Total (i.e. Null); 397 Residual
```

```
## Null Deviance: 500
```

```
## Residual Deviance: 480.3 AIC: 486.3
```

```
...
```

Suppose our model has p parameters (the number of β s, including the intercept). Then the AIC is

$$AIC = \text{deviance} + 2p$$

AIC

$$AIC = \text{deviance} + 2p$$

Model 1:

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \textit{Term}_i + \beta_2 \log(\textit{DisbursementGross}_i)$$

Residual Deviance: 3974 AIC: 3980

Model 2:

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \textit{Term}_i + \beta_2 \textit{Urban}_i + \beta_3 \textit{Rural}_i + \beta_4 \textit{Employees}_i$$

Residual Deviance: 3827 AIC:

Which model do we prefer, based on AIC?

Back to the research question

Research question: Which combination of variables "best" models loan default?

We need:

- + A metric to compare different models
 - + Solution: AIC
- + A way to efficiently search through many different models

How should we search many different models?

Some model search algorithms

Best subset selection:

Stepwise selection:

Class activity

https://sta214-f25.github.io/class_activities/ca_11.html

- + Work independently or with a neighbor
- + Submit HTML on Canvas when done